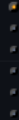


State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts —
Based on 23,262 developers worldwide

RESPONDENTS	COVERAGE	PERIOD
23,262 developers	Languages · Tools · AI · Careers	2017 - 2024 longitudinal
AUDIENCE	Engineering Leaders · CTOs · DevEx Teams	



Python Closes the Gap: 8 Years of Usage Shifts

LANGUAGE	2017	2020	2022	2024	Δ 17→24	8-YEAR TREND
JavaScript	65%	70%	65%	61%	▼ -4 pp	
Python	32%	55%	53%	57%	▲ +25 pp	
HTML/CSS	60%	61%	54%	51%	▼ -9 pp	
SQL	42%	56%	49%	48%	● +6 pp	
Java	47%	54%	48%	46%	▼ -1 pp	
TypeScript	12%	28%	34%	37%	▲ +25 pp	
Go	8%	19%	19%	18%	● +10 pp	
Rust	—	7%	9%	11%	▲ n/a→11%	
PHP	30%	27%	20%	17%	▼ -13 pp	
Kotlin	2%	17%	16%	14%	● +12 pp	

▲ Python +25 pp (2017→2024)

▲ TypeScript tripled: 12% → 37%

▼ PHP -13 pp long-term decline

● JS peaked at 70% in 2020, now 61%

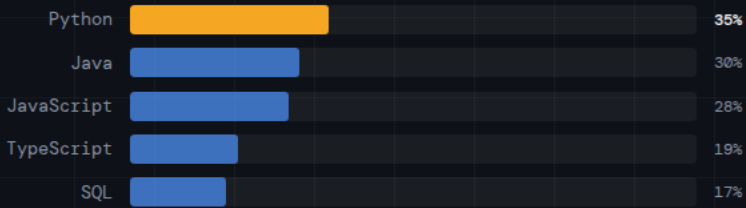
Python Is Now the #1 Primary Language at 35%

Ecosystem lock-in is real. Adoption intent stays below 6% across all languages — developers rarely switch primary stacks once committed.

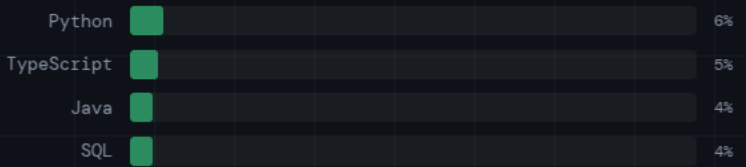
Python overtakes JS as primary

Max intent: only 6% plan to adopt Python

PRIMARY LANGUAGE — % OF DEVELOPERS



ADOPTION INTENT — % PLANNING TO ADOPT AS PRIMARY



TypeScript & Rust Top Satisfaction, Nearly Tied

199

TypeScript score vs Rust at 198

Type safety + modern tooling = happier developers. Both TypeScript and Rust prioritize expressive type systems and compile-time safety guarantees.

TypeScript #1 at 199

Rust #2 at 198 (only 1 pt gap)

SATISFACTION SCORE – TOP 10 LANGUAGES (HIGHER = MORE LOVED)



Each Language Owns a Domain: AI → Python, UI → TypeScript

APPLICATION DOMAIN	LEADING LANGUAGE	SHARE	SECONDARY LANGUAGES
AI / Machine Learning	Python	31%	R, Julia, C++
Data Science & Analytics	Python	28%	SQL, R
User Interface (Frontend)	TypeScript	62%	JavaScript (58%)
APIs & Services	Java	49%	Python (38%), Go
Application Logic & Workflows	Java	55%	Python, Kotlin
System Tools & Components	C++	27%	C, Rust, Go
Data Processing & Analytics	Python	17%	SQL, R, Spark
Hardware Interfacing	C++	21%	C, Rust

Python: dominant across 3+ data/AI domains

TypeScript: 62% UI share – highest single-domain share

Java: owns enterprise backend (APIs + application logic)

C++: irreplaceable in hardware & systems

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML Teams

Python is the **undisputed leader** for AI (31%), data science (28%), and data processing — and growing. Expand Python hiring and toolchain investment for any AI initiative.

Action: hire & upskill

02

TypeScript Is the Future of Frontend & Full-Stack

TypeScript holds **62% of UI development** and the #1 satisfaction score (199). Standardize on TypeScript — it has both market dominance and developer love.

Action: standardize

03

Rust & Go Are the Infrastructure Growth Plays

Go stabilized at 18% after rapid growth; Rust is climbing steadily to 11%. Both excel for **cloud-native, microservices, and systems work** where performance is non-negotiable.

Action: strategic bets

04

PHP & Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to **17% over 7 years**. Ruby fell from 10% to 4%. Plan for reduced talent supply and build migration paths toward modern alternatives.

Action: plan migration

05

Developer Satisfaction Correlates with Type Safety — Factor This Into Retention

TypeScript (199) and Rust (198) lead satisfaction because they offer **type safety + modern tooling + expressive systems**. Developers rarely switch ecosystems — so initial language choice drives long-term team happiness and retention. Prioritize languages with high satisfaction when building new teams.

Action: retention & team design